



## 第四章 频域分析法



### 复习

#### ● 频率特性 $G(j\omega)$ 的定义

➤ 定义一:  $G(j\omega) = |G(j\omega)| \angle G(j\omega)$

➤ 定义二:  $G(j\omega) = G(s)|_{s=j\omega}$

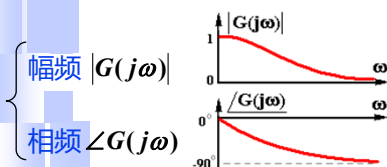
➤ 定义三:  $G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$



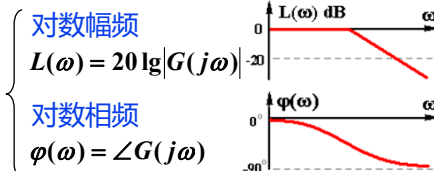
## 复习

### ● 频率特性 $G(j\omega)$ 的表示方法

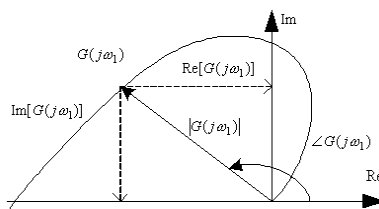
#### I. 频率特性



#### II. 对数频率特性(Bode)



#### III. 幅相特性(Nyquist)



## 复习

### (6) 振荡环节

#### ● 谐振频率 $\omega_r$ 和谐振峰值 $M_r$

$$\frac{\omega^2}{\omega_n^2} = 1 - 2\xi^2$$

$$\omega_r = \omega_n \sqrt{1 - 2\xi^2}$$

$$M_r = |G(j\omega_r)| = \frac{1}{2\xi\sqrt{1 - \xi^2}}$$

$$\xi > 0.707 \quad (\beta < 45^\circ)$$

$$1 - 2\xi^2 < 0 \quad \omega_r, M_r \text{ 不存在}$$

$$\xi = 0.707 \quad (\beta = 45^\circ)$$

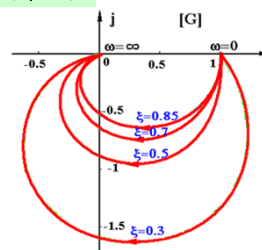
$$1 - 2\xi^2 = 0 \quad \begin{cases} \omega_r = 0 \\ M_r = 1 \end{cases}$$

$$0 < \xi < 0.707 \quad (45^\circ < \beta < 90^\circ)$$

$$1 - 2\xi^2 > 0 \quad \begin{cases} \omega_r = \omega_n \sqrt{1 - 2\xi^2} < \omega_n \\ M_r = \frac{1}{2\xi\sqrt{1 - \xi^2}} > 1 \end{cases}$$

$$\xi = 0 \quad (\beta = 90^\circ)$$

$$1 - 2\xi^2 = 1 \quad \begin{cases} \omega_r = \omega_n \\ M_r = \infty \end{cases}$$

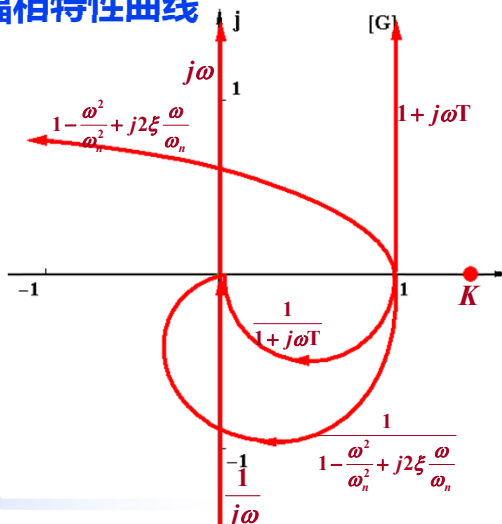




## 复习

### ● 典型环节的幅相特性曲线

- (1)  $G(j\omega) = K$
- (2)  $G(j\omega) = j\omega$
- (3)  $G(j\omega) = 1/j\omega$
- (4)  $G(j\omega) = 1/(1 + j\omega T)$
- (5)  $G(j\omega) = 1 + j\omega T$
- (6)  $G(j\omega) = 1/\left(1 - \frac{\omega^2}{\omega_n^2} + j2\xi \frac{\omega}{\omega_n}\right)$
- (7)  $G(j\omega) = 1 - \frac{\omega^2}{\omega_n^2} + j2\xi \frac{\omega}{\omega_n}$



## 幅相频率特性

### ● 开环幅相特性曲线的绘制

$$G(s) = \frac{K(\tau_1 s + 1) \cdots (\tau_m s + 1)}{s^v (T_1 s + 1) \cdots (T_{n-v} s + 1)} = \frac{K \prod_{i=1}^m (\tau_i s + 1)}{s^v \prod_{j=1}^{n-v} (T_j s + 1)}$$

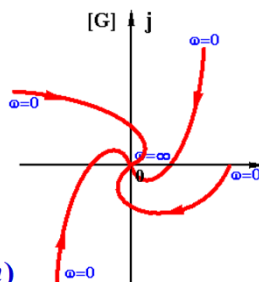
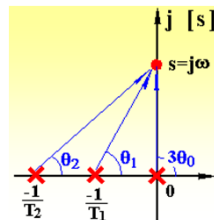
$$\left\{ \begin{aligned} |G(\omega)| &= \frac{K \prod_{i=1}^m |1 + j\tau_i \omega|}{|\omega|^v \prod_{j=1}^{n-v} |1 + jT_j \omega|} = \frac{K \prod_{i=1}^m \sqrt{1 + \tau_i^2 \omega^2}}{\omega^v \prod_{j=1}^{n-v} \sqrt{1 + T_j^2 \omega^2}} \\ \varphi(\omega) &= \angle G(j\omega) = \sum_{i=1}^m \angle(1 + j\tau_i \omega) - v \times 90^\circ - \sum_{j=1}^{n-v} \angle(1 + jT_j \omega) \\ &= \arctan \tau_1 \omega + \cdots + \arctan \tau_m \omega \\ &\quad - v \times 90^\circ - \arctan T_1 \omega - \cdots - \arctan T_{n-v} \omega \end{aligned} \right.$$



## 幅相频率特性

例:  $G(s) = \frac{K}{s^v (T_1 s + 1)(T_2 s + 1)} = \frac{K/(T_1 T_2)}{s^v (s + 1/T_1)(s + 1/T_2)}$

| $v$ | $G(j\omega)$                                              | $G(j0)$                    | $G(j\infty)$          |
|-----|-----------------------------------------------------------|----------------------------|-----------------------|
| 0   | $\frac{K}{(1 + j\omega T_1)(1 + j\omega T_2)}$            | $K \angle 0^\circ$         | $0 \angle -180^\circ$ |
| I   | $\frac{K}{j\omega(1 + j\omega T_1)(1 + j\omega T_2)}$     | $\infty \angle -90^\circ$  | $0 \angle -270^\circ$ |
| II  | $\frac{K}{(j\omega)^2(1 + j\omega T_1)(1 + j\omega T_2)}$ | $\infty \angle -180^\circ$ | $0 \angle -360^\circ$ |
| III | $\frac{K}{(j\omega)^3(1 + j\omega T_1)(1 + j\omega T_2)}$ | $\infty \angle -270^\circ$ | $0 \angle -450^\circ$ |



起点  $\begin{cases} K \angle 0^\circ & v = 0 \\ \infty \angle -90^\circ v & v > 0 \end{cases}$  终点  $0 \angle -90^\circ(n-m)$



## 幅相频率特性

例:  $G_1(s) = \frac{K}{s^2(T_1 s + 1)(T_2 s + 1)}$

$G_1(j0) = \infty \angle -180^\circ$

$\downarrow |G_1| \downarrow \angle G_1 \downarrow$

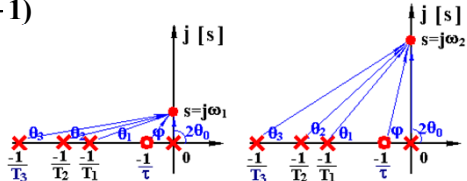
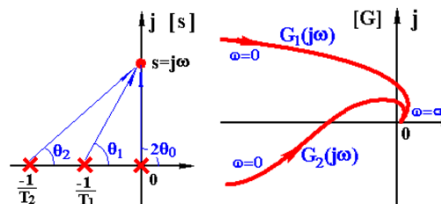
$G_1(j\infty) = 0 \angle -360^\circ$

$G_2(s) = \frac{K(\tau s + 1)}{s^2(T_1 s + 1)(T_2 s + 1)(T_3 s + 1)}$

$G_2(j0) = \infty \angle -180^\circ$

$\downarrow$

$G_2(j\infty) = 0 \angle -360^\circ$





## 幅相频率特性

例:  $G(s) = \frac{s^3}{(s+0.2)(s+1)(s+5)} = \frac{s^3}{(1+5s)(1+s)(1+0.2s)}$

$$G(j0) = 0 \angle +270^\circ$$

$$\downarrow |G| \uparrow \quad \angle G \downarrow$$

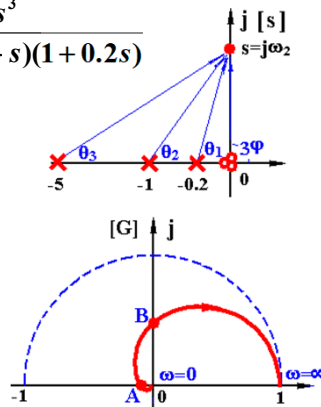
$$G(j\infty) = 1 \angle 0^\circ$$

$$G(j\omega) = \frac{-j\omega^3(1-j5\omega)(1-j\omega)(1-j0.2\omega)}{(1+25\omega^2)(1+\omega^2)(1+0.04\omega^2)}$$

$$= \frac{-\omega^4(6.2-\omega^2)-j\omega^3(1-6.2\omega^2)}{(1+25\omega^2)(1+\omega^2)(1+0.04\omega^2)}$$

$$= X + jY$$

$$A: \begin{cases} \omega_A = 1/\sqrt{6.2} = 0.402 \\ G(j\omega_A) = -0.0267 + j0 \end{cases} \quad B: \begin{cases} \omega_B = \sqrt{6.2} = 2.49 \\ G(j\omega_B) = 0 + j0.412 \end{cases}$$



## 幅相频率特性

例:  $G(s) = \frac{5}{s(s+1)(2s+1)}$

$$G(j\omega) = \frac{5}{j\omega(1+j\omega)(1+j2\omega)} = \frac{-j5(1-j\omega)(1-j2\omega)}{\omega(1+\omega^2)(1+4\omega^2)}$$

$$= \frac{-15}{(1+\omega^2)(1+4\omega^2)} - j \frac{5(1-2\omega^2)}{\omega(1+\omega^2)(1+4\omega^2)}$$

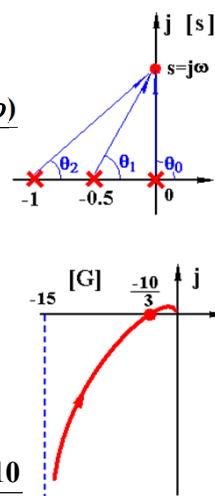
$$G(j0) = \infty \angle -90^\circ$$

$$G(j\infty) = 0 \angle -270^\circ$$

渐近线:  $\text{Re}[G(j0)] \Rightarrow -15$

与实轴交点:  $\text{Im}[G(j\omega)] = 0 \Rightarrow \omega = 1/\sqrt{2} = 0.707$

$$\text{Re}[G(j0.707)] = \frac{-15}{(1+0.5)(1+4 \times 0.5)} = -\frac{10}{3}$$





## 提 纲

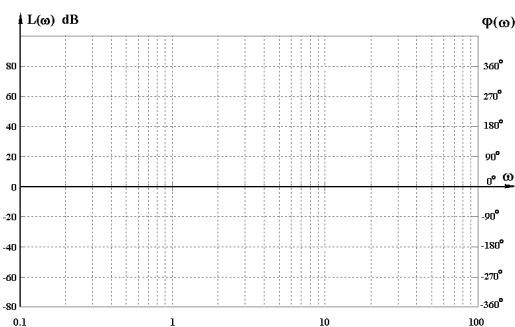
### ● 频域分析法

- 频率特性的基本概念
- 幅相频率特性 (Nyquist图)
- 对数频率特性 (Bode图)
- 频域稳定判据
- 闭环系统的频率特性
- 系统校正



## 对数频率特性

### ● Bode图介绍



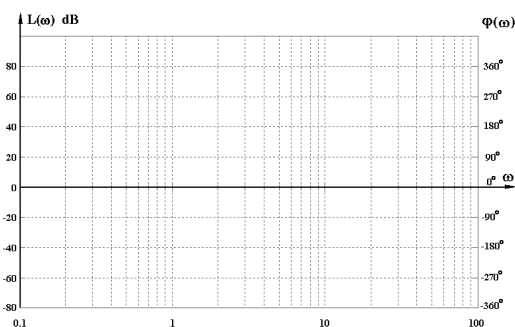
横轴 { 按  $\lg \omega$  刻度, dec “十倍频程” 或 “旬距”  
按  $\omega$  标定, 等距等比

纵轴  $L(\omega) = 20 \lg |G(j\omega)|$  dB “分贝”



## 对数频率特性

### ● Bode图介绍



① 串联环节总的对数幅频特性:  $L(\omega) = L_1(\omega) + \dots + L_n(\omega)$

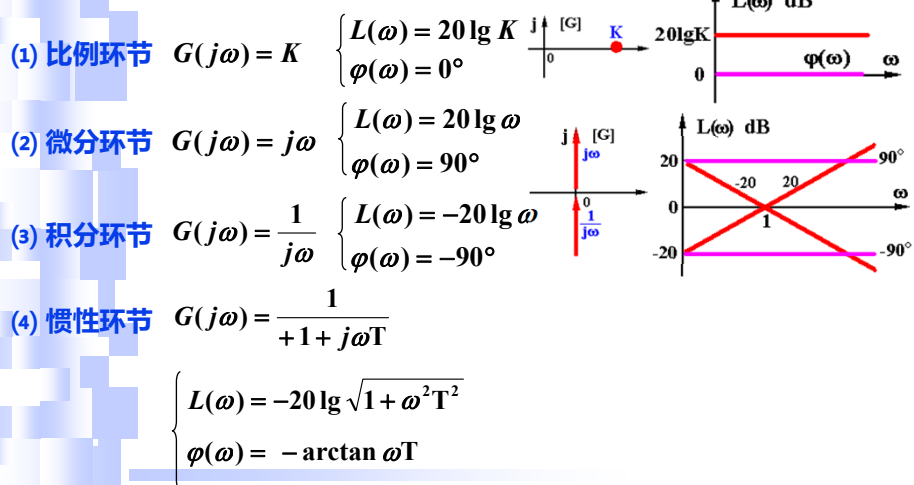
串联环节总的相频特性:  $\varphi(\omega) = \varphi_1(\omega) + \dots + \varphi_n(\omega)$

② 互为倒数的传递函数, 其  $L(\omega)$ 、 $\varphi(\omega)$  均以横坐标成镜像对称



## 对数频率特性

### ● 典型环节的Bode图



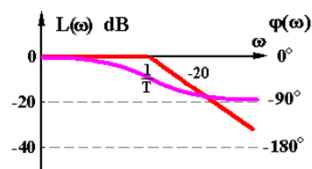
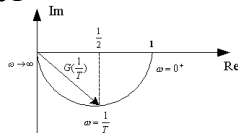


## 对数频率特性

### ● 典型环节的Bode图

(4) 惯性环节  $G(j\omega) = \frac{1}{1 + j\omega T}$

$$\begin{cases} L(\omega) = -20 \lg \sqrt{1 + \omega^2 T^2} \\ \varphi(\omega) = -\arctan \omega T \end{cases}$$



$$w \ll \frac{1}{T}, L(w) \approx -20 \lg 1 = 0$$

$$w \gg \frac{1}{T}, L(w) \approx -20 \lg wT = -20 \lg w - 20 \lg T$$

$$\varphi(j0) = 0, \varphi\left(j\frac{1}{T}\right) = -45^\circ, \varphi(j\infty) = -90^\circ$$

惯性环节对数相频特性关于( $w=1/T, \varphi=-45^\circ$ ) 点对称



## 对数频率特性

### ● 典型环节的Bode图

#### (5) 一阶复合微分

$$G(s) = Ts + 1$$

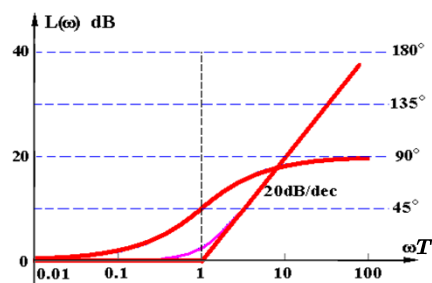
$$G(j\omega) = +1 + j\omega T$$

$$\begin{cases} L(\omega) = 20 \lg \sqrt{1 + \omega^2 T^2} \\ \varphi(\omega) = \arctan \omega T \end{cases}$$

$$w \ll \frac{1}{T}, L(w) \approx 20 \lg 1 = 0$$

$$w \gg \frac{1}{T}, L(w) \approx 20 \lg wT = 20 \lg w + 20 \lg T$$

$$\varphi(j0) = 0, \varphi\left(j\frac{1}{T}\right) = 45^\circ, \varphi(j\infty) = 90^\circ$$







## 对数频率特性

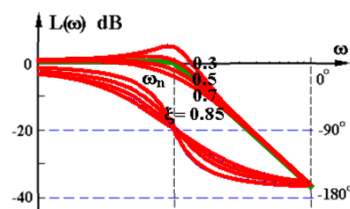
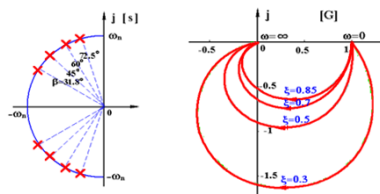
### (6) 振荡环节

$$G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$G(j\omega) = \frac{1}{1 - \frac{\omega^2}{\omega_n^2} + j2\xi\frac{\omega}{\omega_n}}$$

$$\begin{cases} L(\omega) = -20 \lg \sqrt{1 - \frac{\omega^2}{\omega_n^2} + [2\xi\frac{\omega}{\omega_n}]^2} \\ \varphi(\omega) = -\arctan \left[ \frac{2\xi\frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}} \right] \end{cases}$$

$$\begin{cases} \frac{\omega}{\omega_n} \ll 1 & \begin{cases} L(\omega) \approx 0 \\ \varphi(\omega) \approx 0^\circ \end{cases} \\ \frac{\omega}{\omega_n} \gg 1 & \begin{cases} L(\omega) \approx -40 \lg(\omega/\omega_n) \\ \varphi(\omega) \approx -180^\circ \end{cases} \end{cases}$$



## 对数频率特性

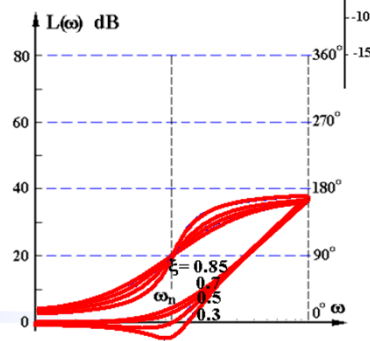
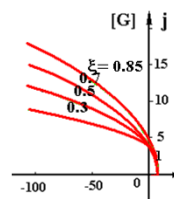
### (7) 二阶复合微分

$$G(s) = \left(\frac{s}{\omega_n}\right)^2 + 2\xi\frac{s}{\omega_n} + 1$$

$$G(j\omega) = 1 - \frac{\omega^2}{\omega_n^2} + j2\xi\frac{\omega}{\omega_n}$$

$$L(\omega) = 20 \lg \sqrt{1 - \frac{\omega^2}{\omega_n^2} + [2\xi\frac{\omega}{\omega_n}]^2}$$

$$\varphi(\omega) = \arctan \frac{2\xi\frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}$$





## 对数频率特性

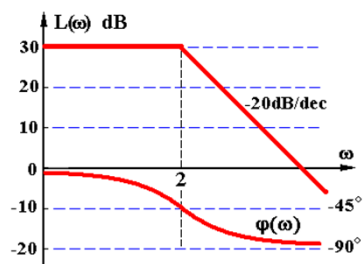
例：根据Bode图确定系统传递函数

解：依图有  $G(s) = \frac{K}{Ts+1}$   
 $20 \lg K = 30 \Rightarrow K = 10^{\frac{30}{20}} = 31.6$

转折频率  $\omega = 2 = 1/T$

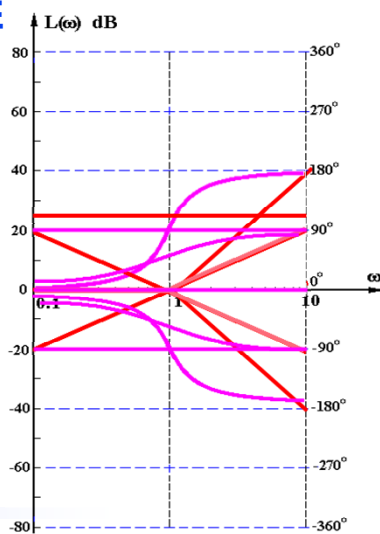
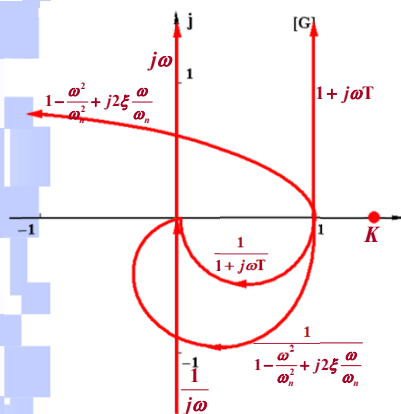
$$T = 0.5$$

$$G(s) = \frac{31.6}{\frac{s}{2} + 1}$$



## 对数频率特性

### ● 典型环节的频率特性





## 对数频率特性

### ● 系统的开环Bode图

$$G(s) = \frac{K(\tau_1 s + 1) \cdots (\tau_m s + 1)}{s^v (T_1 s + 1) \cdots (T_{n-v} s + 1)}$$

$$\left\{ \begin{aligned} L(\omega) &= 20 \lg |G| = 20 \lg \frac{K |1 + j\tau_1 \omega| \cdots |1 + j\tau_m \omega|}{|\omega|^v |1 + jT_1 \omega| \cdots |1 + jT_{n-v} \omega|} \\ &= 20 \lg K + 20 \lg |1 + j\tau_1 \omega| + \cdots + 20 \lg |1 + j\tau_m \omega| \\ &\quad - 20v \lg |\omega| - 20 \lg |1 + jT_1 \omega| - \cdots - 20 \lg |1 + jT_{n-v} \omega| \\ \varphi(\omega) &= \angle G \\ &= \arctan \tau_1 \omega + \cdots + \arctan \tau_m \omega \\ &\quad - 90^\circ v - \arctan T_1 \omega - \cdots - \arctan T_{n-v} \omega \end{aligned} \right.$$



## 对数频率特性

$$G(s) = \frac{K(\tau_1 s + 1) \cdots (\tau_m s + 1)}{s^v (T_1 s + 1) \cdots (T_{n-v} s + 1)}$$

### ● 绘制系统开环Bode图的步骤

例  $G(s) = \frac{40(s + 0.5)}{s(s + 0.2)(s^2 + s + 1)}$

$$G(s) = \frac{100(\frac{s}{0.5} + 1)}{s(\frac{s}{0.2} + 1)(s^2 + s + 1)}$$

(1) 化G(s)为尾1标准型

(2) 顺序列出转折频率

$$G(j\omega) \approx \frac{K}{(j\omega)^v}$$

$$L \approx 20 \lg K - 20v \lg \omega$$

(3) 确定基准线

最小转折频率之左的特性及其延长线

(4) 叠加作图

一阶 { 惯性环节 -20dB/dec  
复合微分 +20dB/dec  
二阶 { 振荡环节 -40dB/dec  
复合微分 +40dB/dec

{ 0.2 惯性环节  
0.5 一阶复合微分  
1 振荡环节

{ 基准点 ( $\omega = 1, L(1) = 20 \lg K$ )  
( $\omega = K^{1/v}, L(K^{1/v}) = 0$ )  
斜率  $-20 \cdot v \text{ dB/dec}$

{ w=0.2 惯性环节 -20  
w=0.5 一阶复合微分 +20  
w=1 振荡环节 -40



## 对数频率特

$$G(s) = \frac{100(\frac{s}{0.5} + 1)}{s(\frac{s}{0.2} + 1)(s^2 + s + 1)}$$

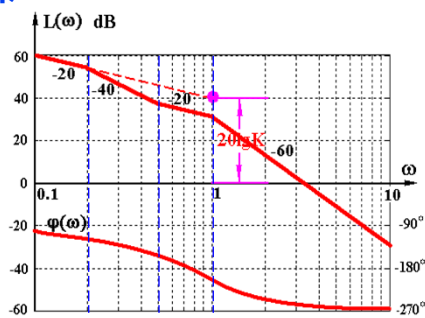
### ● 绘制系统开环Bode图的步骤

基准点 ( $\omega = 1$ ,  $L(1) = 20 \lg K$ )  
斜率  $-20 \cdot \nu$  dB/dec

$\begin{cases} \omega=0.2 \text{ 惯性环节} & -20 \\ \omega=0.5 \text{ 一阶复合微分} & +20 \\ \omega=1 \text{ 振荡环节} & -40 \end{cases}$

(5) 修正  $\begin{cases} \text{① 两惯性环节转折频率很接近时} \\ \text{② 振荡环节 } \varepsilon \notin (0.38, 0.8) \text{ 时} \end{cases}$

(6) 检查  $\begin{cases} \text{① } L(\omega) \text{ 最右端曲线斜率} = -20(n-m) \text{ dB/dec} \\ \text{② 转折点数} = (\text{惯性}) + (\text{一阶复合微分}) + (\text{振荡}) + (\text{二阶复合微分}) \\ \text{③ } j(\omega) \Rightarrow -90^\circ(n-m) \end{cases}$



## 对数频率特性

● 例:  $G(s) = \frac{s^3}{(s+0.2)(s+1)(s+5)}$ , 绘制Bode图。

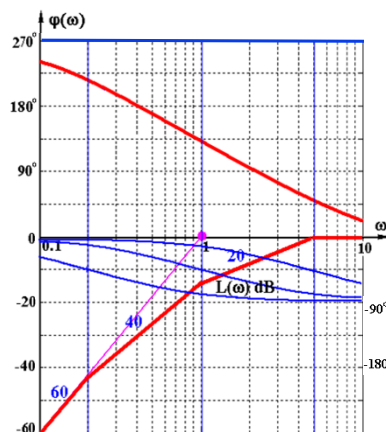
解 ① 标准型  $G(s) = \frac{s^3}{(\frac{s}{0.2} + 1)(s+1)(\frac{s}{5} + 1)}$

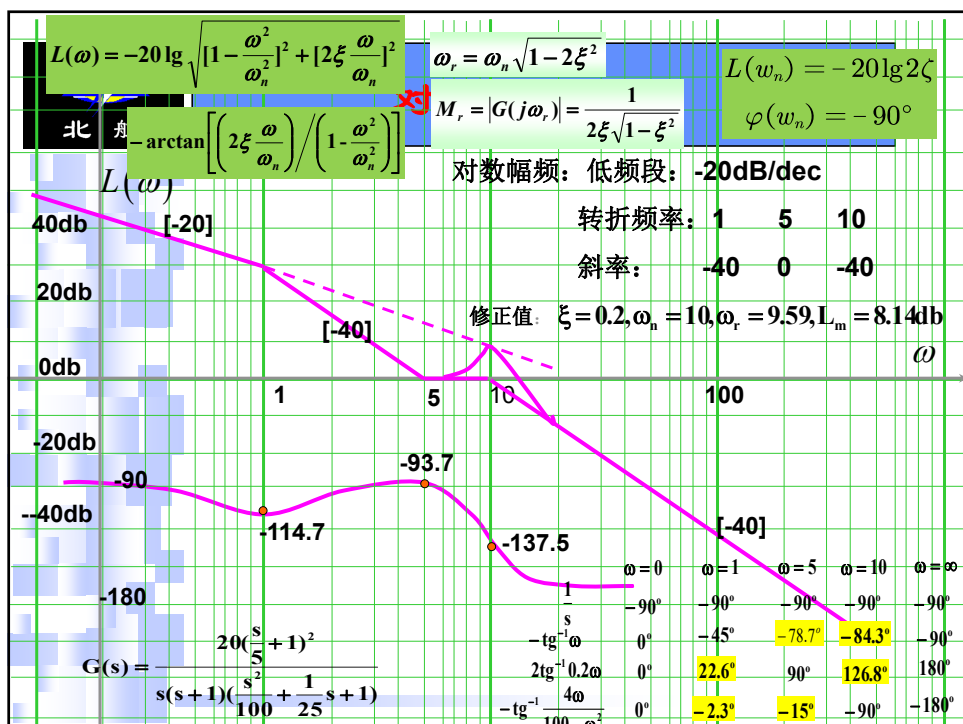
② 转折频率  $\begin{cases} \omega_1 = 0.2 \Rightarrow -20 \\ \omega_2 = 1 \Rightarrow -20 \\ \omega_3 = 5 \Rightarrow -20 \end{cases}$

③ 基准线  $\begin{cases} \text{基点 } (\omega = 1, 20 \lg 1 = 0 \text{ dB}) \\ \text{斜率 } -20 \times (-3) = 60 \text{ dB/dec} \end{cases}$

④ 作图

⑤ 检查  $\begin{cases} L(\omega) \text{ 最右端斜率} = -20(n-m) = 0 \\ \text{转折点数} = 3 \\ j(\omega) \text{ 最终趋于 } -90^\circ(n-m) = 0^\circ \end{cases}$





## 实验二

- 资料：MWORKS软件安装、常用命令、实验指导书、实验APP使用说明

<https://bhpan.buaa.edu.cn/link/AA0D0B6211E27448449EBA5EDA3DCE73D1>

文件夹名：实验指导资料

- 实验报告提交网盘：

<https://bhpan.buaa.edu.cn/link/AAEF72EC7EF6114556A9F8E995DDB4FFD0>

文件夹名：实验二

有效期限：2026-04-26 23:59